

EVALUATION OF TWO METHODS TO COLLECT CANINE MELANOCYTES



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Abstract

Introduction. One of the most difficult aspects of employing precision medicine in the treatment of cancer is determining the best targets to treat within the tumor. A trademark of cancers is that while two patients may present similar tumors of the same tissue, even at the same stages of development, they will likely not respond in the same way to identical treatment. This issue is common throughout all types of cancers and presents a significant hurdle in creating effective treatment regimens for patients. Melanoma is a significant tumor of dogs that typically occurs in the oral mucosa. Normal melanocytes from canine melanoma patients are needed to compare transcriptomes to patient melanomas using deep sequencing and from this data, therapeutic targets may be selected. The current project seeks to identify and obtain appropriate normal melanocytes from patient animals with minimal additional morbidity. Epidermal melanocytes represent a population of approximately 3-7% of a cell types throughout the skin, making them a difficult target to isolate. Herein, we evaluate two potential methods for the isolation of pure canine melanocytes from normal skin.

Methods. The first method is a differential adhesion method using cell culture. Briefly; punch biopsies from normal canine skin were incubated overnight in dispase. After incubation the epidermis was separated from the dermis and treated with trypsin EDTA or 1x TrypLE recombinant trypsin to disperse the epidermal cells into a single cell suspension and cultured in Melanocyte Growth Medium M2.

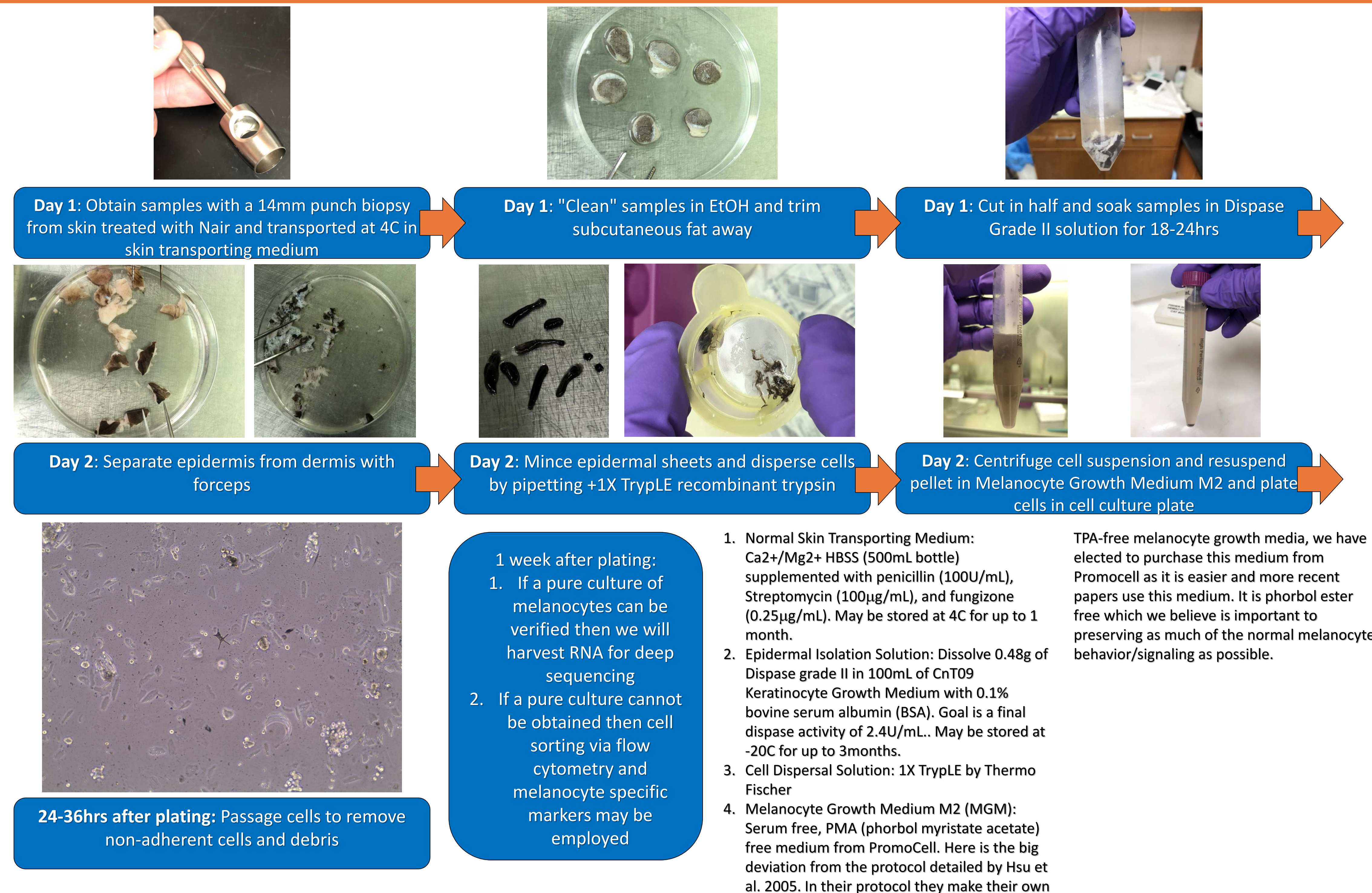
Method two involves the plucking of anagen hair follicles from normal dog skin for explant culture. Approximately 80-100 hairs were plucked from the craniodorsal region of healthy dogs under non-sterile conditions and 30-50 anagen hairs with attached follicles were selected under a microscope. The hairs were washed in melanocyte growth media before explanting 12-15 follicles onto each microporous membrane in Melanocyte Growth Medium. Cells were cultured under normal cell culture conditions.

Results. Method one resulted in the isolation of identifiable melanocytes from non-haired oral mucosa. Thus far pure cultures of melanocytes from haired skin have not been isolated by this method. Method two has also not resulted in the isolation of identifiable melanocytes.

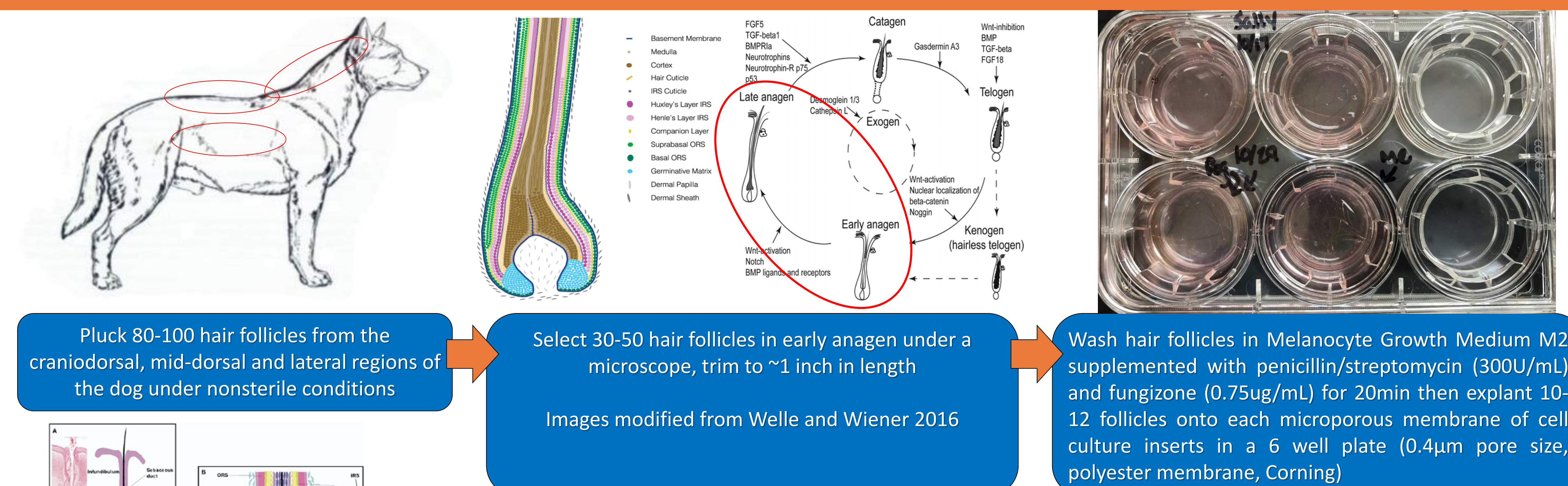
Conclusions. The isolation of melanocytes from oral mucosa, while possible, is not practical in patients already undergoing significant oral resection for tumor. Unfortunately, published methods of isolating melanocytes from haired regions of the dog, either by punch biopsy or follicle isolation, has not yet proven feasible. The largest issue to date is that hair follicles act as anchors for the epidermis in normal canine skin making separation from the dermis difficult. We are employing different alterations to attempt to overcome this hurdle as well as looking into alternate isolation methods

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Normal Skin Protocol

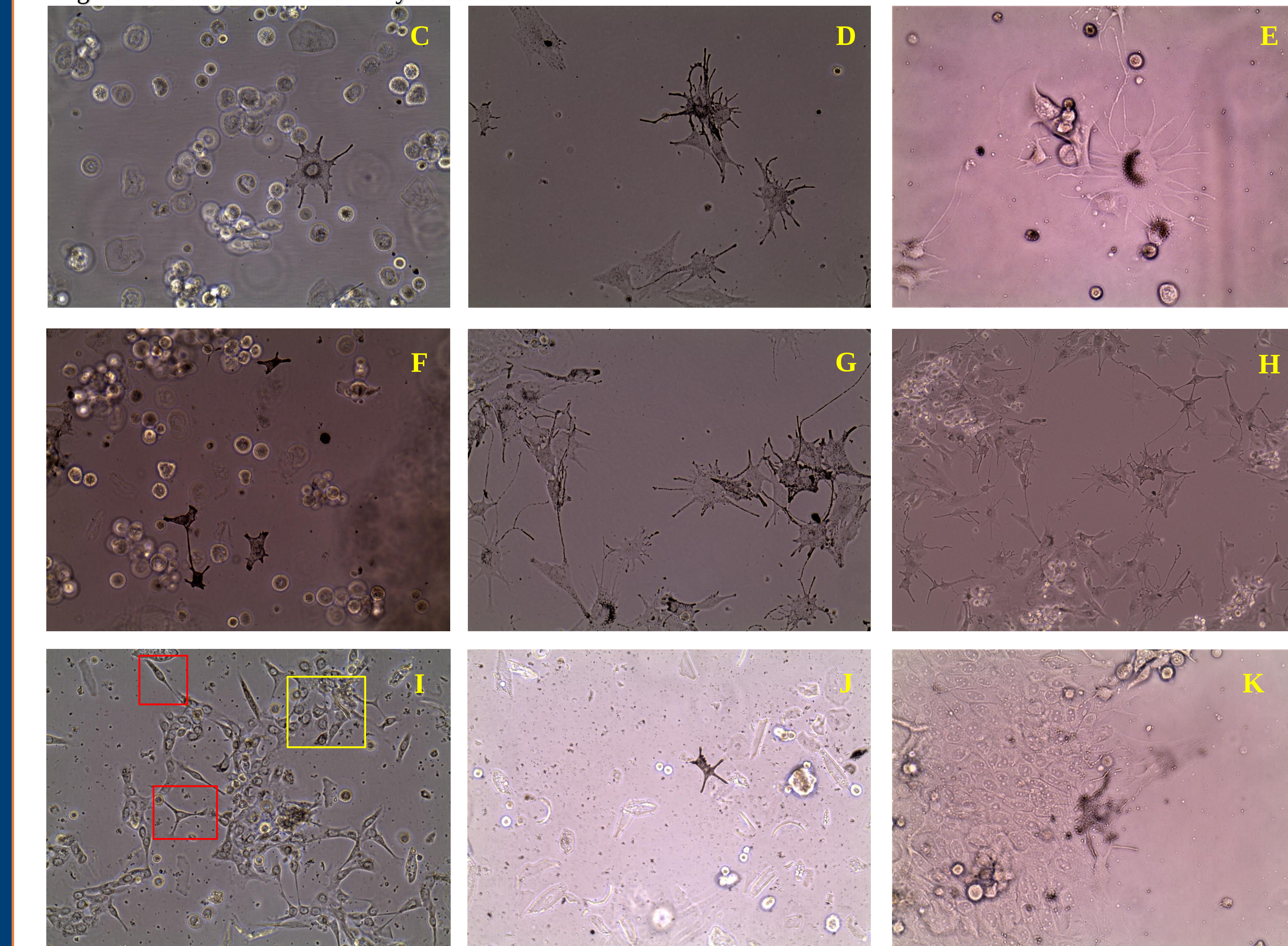


Hair Follicle Protocol



Results

Figure A: Anagen hair follicle seen at 20X
Figure B: Anagen hair follicle seen at 40X
Figure C: Oral mucosa melanocyte seen at 40X
Figure D: Oral mucosa melanocytes seen at 40X
Figure E: Unconfirmed melanocyte seen at 40X
Figure F: Oral mucosa melanocytes seen at 20X
Figure G: Oral mucosa melanocytes seen at 40X
Figure H: Oral mucosa melanocytes seen at 20X
Figure I: Skin melanocytes (red box) and keratinocytes (yellow box) seen at 40X
Figure J: Skin melanocyte seen at 40X
Figure K: Oral mucosa keratinocytes seen at 40X



Conclusions

Normal Skin:

- Being extremely haired, normal skin presents a unique challenge in the separation of the epidermis and dermis, but this may be overcome through the use of Nair and maximizing sample exposure to Dispace by incubating samples in a petri dish instead of a tube.
- This protocol has been proven to be effective in oral mucosa samples, however the challenges presented by haired skin make for a difficult process that results in extremely low yield of melanocytes and has yet to yield a pure melanocyte culture, as has been done in previous studies using human fetal foreskin.
- As a result, we are now looking at alternative methods for obtaining melanocytes, such as simply homogenizing the whole biopsy and using melanocyte specific antibodies for physical cell sorting of melanocytes by flow cytometry OR single cell sequencing of tumor margins to distinguish normal melanocytes from cancerous melanocytes.

Hair Follicle Plucking:

- We have yet to obtain any melanocytes from canine hair follicles, however a previous study in human hair showed this to be a very slow process so we are still hopeful.
- An adaption we plan to try is culturing the hair follicles in 5% O₂, 5% CO₂ and 90% N₂. These conditions are generally considered "hypoxic" but actually represent normoxic conditions within the skin *in vivo*.

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